00: The Staging, the Professor, and the Tribunal of Paper 001

AuthorityEffect: None.
Location: Moscow, Sadovaya 302-bis, Apartment 50.
Presiding Tribunal: Messire Woland, Lev Davidovich Landau ("Dau"), Koroviev-Fagotto, Behemoth, Azazello, Abadonna, Hella; with the Russian-American Delegation: Jeremy, Liudmila, Polina, Yaroslava.
Subject of Examination: Paper 001 — *The Symbol Game: Discrete Cochain Retraction, Primal-Dual Obstruction, and Toy Involutions* (PFP I: Exact Graph Repair and the Boundary Between a Proposal and a Certificate).
Companion Documents: - Study Guide: study_guide.md - Examination Paper: quiz.md - Instructor Key: instructor_key.md - Notebook worksheet: paper001_notebook_learning_worksheet.md

The worksheet has six exercises on the notebooks and the Lean sources. It does not replace the thirteen-problem examination.

This is an imagined literary examination. The dialogue and tribunal are fictional, rather than historical quotations or biographical evidence. The mathematical definitions follow the manuscript.

1. The Knock at the Door: The Second Trial

The sleet over Sadovaya Street had thickened into a foul, freezing slush that plastered the bare lime trees along the Garden Ring. Inside Apartment 50, the atmosphere had shifted from the declared two-mark carrier of Paper 000 into the sharp, metallic tension of a computational workshop.

You had practiced the two-mark exercises. Their labels do not establish a verified transition from $S(0)$ to $S(4)$. But now your leather portfolio bulged with printouts from gradient-descent packages, floating-point loss curves, continuous Laplace-Beltrami operators, and PyTorch autograd scripts. On your scratchpad, you had scribbled the fatal, lazy claim:

"A candidate potential with cost 3.0000001 converges to the global minimum."

You knocked on the black oak door.

The door flew back instantly. **Hella** stood before you, holding a brass caliper in one bloodless hand. Her cold green eyes flicked to your satchel:

> *"PyTorch? SciPy `optimize.minimize`? Floating-point epsilon approximations?*
> *Throw those contaminated papers down the stairwell this instant. In*

Paper 001, we calculate in exact rationals $\mathbb{Q}$. If you attempt to smuggle an IEEE 754 float across this threshold, I will sever your tendons with this caliper."

She stepped aside, gesturing with a contemptuous jerk of her chin toward the candlelit salon.

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2. Woland and Jeremy Carroll

In the center of the drawing room sat **Messire Woland**, leaning on his silver poodle-headed cane. His black eye was fixed upon a large sheet of butcher paper spread across the dining table, upon which was drawn a simple directed triangle:

$$a \to b \quad (w_1 = 1, c_1 = 1), \quad b \to c \quad (w_2 = 2, c_2 = 1), \quad c \to a \quad (w_3 = 3, c_3 = 1)$$

Beside Woland stood **Jeremy Carroll**, the author of Paper 001. Jeremy had his sleeves rolled to his elbows, chalk under his fingernails, and a worn mechanical pencil tucked behind his ear:

> *"Look at him, Messire,"* Jeremy said in a dry, calm Michigan drawl—a true Yooper from Sault Ste. Marie who didn't waste syllables. *"He walked all the way from Cambridge or Silicon Valley with twenty thousand dollars worth of GPU cloud credits, and he thinks he can optimize a three-edge graph by running gradient descent.
>
> He doesn't know the difference between a **proposal** and a **certificate**. He thinks that because he found an assignment of node levels that costs three, he has certified that no assignment costs two. He has never seen a dual circulation in his life."*

Woland turned his mismatched eyes upon you. His green eye flared with cold, ironic mirth:

> *"He believes in algorithms, Jeremy. He believes that if an artificial brain sweats through forty iterations of gradient descent, the resulting floating-point tensor constitutes a mathematical theorem.
>
> Step forward, Lev Davidovich! Show our guest what happens when continuous hand-waving meets discrete graph topology."*

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3. Landau Takes the Slate

Lev Davidovich Landau ("Dau") erupted from behind the chalkboard, his railway chalk spinning in a blur between his fingers. He slammed the chalk

against the slate, producing a shriek that rattled the teacups on Liudmila's tray:

*"Quiet! Let us review the crime!

Three clocks on a wall! The first says b is 1 hour ahead of a. The second says c is 1 hour ahead of b. The third says a is 1 hour ahead of c! You walk around the triangle and you are 3 hours in the future! Can this discrepancy be resolved by assigning consistent times s_a, s_b, s_c?

Of course not! The reports $c = (1, 1, 1)$ do not lie in the image of the discrete coboundary operator B! There is a topological obstruction $\Phi_1(c) > 0$!

And what does our modern student do? He proposes levels $s = (0, -2, -1)$. He calculates the cost:

$$D_c(s) = 1|1-(-2-0)|+2|1-(-1-(-2))|+3|1-(0-(-1))| = 1(3)+2(0)+3(0) = 3$$

And then he proudly writes: 'Q.E.D.! Three is the minimum!'

An absolute scandal! A disgrace to Kharkiv! Proposing a level vector s only gives an **upper bound**: $\Phi_1(c) \leq D_c(s) = 3$. How do you know there does not exist some fractional potential $s' \in \mathbb{Q}^3$ that achieves cost $2\frac{1}{2}$? Where is your lower bound? Where is your dual witness λ? Where is your circulation?

For this examination, exhibit an exact circulation $\lambda \in \mathbb{Q}^E$ satisfying $B^T\lambda = 0$ and $|\lambda_e| \leq w_e$ whose value $\langle \lambda, c \rangle$ matches your cost 3. That is your independently checkable lower bound; failed heuristics cannot supply it!"*

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4. The Retinue and the Delegation

The room was filled with judges, each watching for a specific failure mode:

- **Liudmila (Luda)** poured steaming dark tea from the samovar into a faceted glass, sliding a saucer of wild strawberry jam toward you: > *"Breathe, young man. Don't let Dau give you heart palpitations. Drink your tea and write down the definitions cleanly: the incidence matrix B, the cochain c, the penalty weights w. It is just linear algebra over the rationals. Weak duality is your friend: $\langle \lambda, c \rangle \leq D_c(s)$. Once the two numbers meet, Dau cannot touch you."*
- **Polina** hopped up onto the edge of the dining table, dangling her sneakers and smirking: > *"Dad, ten bucks says he tries to divide by zero when he*

computes the weighted cycle formula $\min_e w_e|\sum c_e|$. *Or worse—he'll forget that edges have signs! Look at his pencil, it's shaking already."*

- **Yaroslava** reset her mechanical stopwatch with a crisp, metallic click: > *"Standard protocol. Three pages. Page 1 is elementary evaluations. Page 2 is the primal-dual furnace. Page 3 is the infinite descent trap and toy involutions. You have exactly forty-five minutes. Begemot, keep that sulfur flame away from the paper."*
- **Koroviev (Fagot)**, perched on his stool in checkered rags, beamed maliciously through his cracked pince-nez: > *"Oh, Messire! Look at his face! He wants to run 100,000 spanning trees in Python! He doesn't know that the native Rust crate* `graph_phi1.rs` *enumerates spanning forests exactly and returns* `EnumerationLimit` *when it would exceed 100,000 complete forests; the count cap is not a runtime guarantee! He thinks computation is free!"*
- **Behemoth**, the giant black cat, sat cross-legged on the table, stirring his tea with a steel compass and gnawing on a salted herring head: > *"If the candidate writes that an odd frustrated cycle can be eliminated by a local gauge transformation, I shall bite his ear off. Parity is invariant, my dear sir! You cannot turn an odd cycle into an even cycle by combing the fur!"*
- **Azazello** leaned against the window jamb, paring his yellow fingernails with an obsidian dagger: > *"One float. Just one* `0.3333`. *That is all I ask. The window is already unlatched."*
- **Abadonna** sat silently in his corner behind his opaque black goggles. If you attempted to pass an unfeasible dual circulation with $|\lambda_e| > w_e$, Abadonna would glance at your paper and turn the ink to lead poison.

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5. The Seven Mortal Sins of the Paper 001 Examinee

Landau strode to the slate and, with rapid, violent cuts of the chalk, carved the **Seven Mortal Sins of Paper 001**:

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            THE SEVEN MORTAL SINS OF THE PAPER 001 EXAMINEE
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1. THE UNCERTIFIED PROPOSAL (CONFUSING UPPER BOUND WITH TRUTH):
   Asserting that a candidate potential s achieves the minimum cost without
   exhibiting an independent dual circulation lambda that certifies the lower bound.
   Penalty: Koroviev reads the candidate's diary aloud in front of the salon.

2. THE FLOATING-POINT SLOP (IEEE 754 SMUGGLING):
   Writing decimals (3.0, 0.5, 1e-6) or using gradient descent approximations.
   All assessed graph calculations use exact rationals Q; drawing conversions are not certi
   Penalty: Automatic score = -infinity; Azazello takes your pen.

3. THE OVER-CAPACITY DUAL (FEASIBILITY CHEATING):
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   Claiming a dual witness lambda that violates edge capacity (|lambda_e| > w_e)
   or fails flow conservation (B^T lambda != 0 at any vertex).
   Penalty: Behemoth overturns the boiling samovar into your lap.

4. THE INFINITE DESCENT CONFUSION (ZENO'S TRAP):
   Believing that strict rational descent (D_{t+1} < D_t) implies finite
   termination. The sequence D_t = 2^{-t} descends strictly forever without
   reaching zero.
   Penalty: Hella locks the examinee in the pantry until sunrise.

5. THE GAUGE INVARIANCE AMNESIA:
   Forgetting that shifting edge reports by an exact coboundary (c -> c + B phi)
   leaves the quotient minimum Phi_1(c) strictly unchanged.
   Penalty: Dau tosses the candidate's notebook out the third-floor window.

6. THE TOY INVOLUTION FRAUD (HYPE OVERREACH):
   Promoting the toy sign involution sigma^2 = id in InvolutionLaws.lean
   into a general solution to the continuous quantum fermion sign problem.
   Penalty: Abadonna lowers his black glasses by 2 millimeters.

7. THE 2-CELL HALLUCINATION (IMAGINARY FACES):
   Writing face boundary operators d_1 or claiming nonzero face curvature
   on a graph with zero filled 2-cells (C^2 = 0).
   Penalty: Summary expulsion to Patriarch's Ponds.
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6. The Examination Rules

Jeremy Carroll tapped his pencil on the table:

> *"Here are the ground rules for Paper 001:
>
> 1. **The Graph:** Vertices V, directed edges E, reports $c \in \mathbb{Q}^E$, penalties $w \in \mathbb{Q}^E_{>0}$.
> 2. **The Residual:** For potential $s \in \mathbb{Q}^V$, $r = c - Bs$. Cost $D_c(s) = \sum_e w_e |r_e|$.
> 3. **The Dual:** Flow $\lambda \in \mathbb{Q}^E$, conservation $B^T\lambda = 0$, capacities $|\lambda_e| \leq w_e$.
> 4. **The Zero Duality Gap:** If the dual is feasible and $\langle \lambda, c \rangle = D_c(s)$, the minimum is certified: $\Phi_1(c) = \langle \lambda, c \rangle = D_c(s)$.
> 5. **Exactness:** Every fraction must be written as p/q with $p, q \in \mathbb{Z}$ in lowest terms."*

Woland raised his silver cane and brought it down with a thunderous strike upon the hardwood floor:

"Turn the page, examinee. Let us see if you can repair a graph without drowning in floating-point fiction."

7. Examiner standard from the short companion

Landau points at the declared carrier. Woland asks which claim its checker can establish. Behemoth asks for the smallest valid near miss. Their task in this companion is **exact rational repair and dual witnesses**.

A lower-case wire token names a role only after its interpretation is declared. Arrows sketch graph edges; they do not automatically attach higher-dimensional cells. A receipt verifies its stated finite check and grants no authority to extend Genesis or make a physical claim.

The examiner's standard is a concrete calculation, an explicit counterexample, or a source-bound missing obligation. The manuscript source reading governs the answer.